

6 Tips pour améliorer sa web app Angular



Camille Pillot

Développeuse web



cpillot@takima.fr



[@CamiPillotDev](https://twitter.com/CamiPillotDev)





takima

Camille Pillot

Développeuse web



cpillot@takima.fr



@CamiPillotDev





Les Alias

Mes imports avant

```
import { Employee } from '../..core/models/employee';  
import { EmployeeService } from '../..core/service/employee.service';  
import { Banner } from '../..../shared/banner';  
import { Mood } from '../models/mood';  
import { MoodService } from '../mood.service';
```

Les alias Solution



tsconfig.ts

Mon tsconfig

```
"compilerOptions": {  
  "outDir": "./dist/out-tsc",  
  "strict": true,  
  "target": "es2017",  
  "module": "es2020",  
  "lib": [ "es2020", "dom" ]  
},
```

Mon tsconfig

```
"compilerOptions": {  
  "baseUrl": ".",  
  "paths": {  
    "@app/*": ["src/app/*"],  
    "@core/*": ["src/core/*"],  
    "@shared/*": ["src/shared/*"],  
  },  
  "outDir": "./dist/out-tsc",  
  "strict": true,  
  "target": "es2017",  
  "module": "es2020",  
  "lib": [ "es2020", "dom" ]  
},
```

Mes imports avant

```
import { Employee } from '../..core/models/employee';  
import { EmployeeService } from '../..core/service/employee.service';  
import { Banner } from '../..../shared/banner';  
import { Mood } from '../models/mood';  
import { MoodService } from '../mood.service';
```

Mes imports après

```
import { Employee } from '../..core/models/employee';  
import { EmployeeService } from '../..core/service/employee.service';  
import { Banner } from '../..shared/banner';  
import { Mood } from '../models/mood';  
import { MoodService } from '../mood.service';
```



```
import { Employee } from '@core/models/employee';  
import { EmployeeService } from '@core/service/employee.service';  
import { Banner } from '@shared/models/banner';  
import { Mood } from '@app/models/mood';  
import { MoodService } from '@app/mood.service';
```

Mes imports après

```
import { Employee } from '../../../core/models/employee';  
import { EmployeeService } from '../../../core/service/employee.service';  
import { Banner } from '../../../../shared/banner';  
import { Mood } from '../models/mood';  
import { MoodService } from '../mood.service';
```



```
import { Employee } from '@core/models/employee';  
import { EmployeeService } from '@core/service/employee.service';  
import { Banner } from '@shared/models/banner';  
import { Mood } from '@app/models/mood';  
import { MoodService } from '@app/mood.service';
```



Mes imports de style avant

```
@import '../.../styles/special-theme/colors.scss';  
@import '../.../styles/variables/variable.scss';  
  
.banner {  
    color: $text-default-color;  
    padding: $large-padding;  
}
```

Les alias Solution



angular.json

Mon angular.json

```
"projects": {  
  "alias": {  
    ...  
    "architect": {  
      "build": {  
        "options": {  
          ...  
          "styles": [  
            "src/styles.scss"  
          ],  
          "scripts": []  
        },  
        ...  
      }  
    }  
  }  
}
```

Mon angular.json

```
"projects": {  
  "alias": {  
    ...  
    "architect": {  
      "build": {  
        "options": {  
          ...  
          "styles": [  
            "src/styles.scss"  
          ],  
          "stylePreprocessorOptions": {  
            "includePaths": [  
              "src/styles/special-theme",  
              "src/styles/variables"  
            ]  
          },  
        },  
        "scripts": []  
      },  
      ...  
    }  
  }  
}
```

Mes imports de style avant

```
@import '../.../styles/special-theme/colors.scss' ;  
@import '../.../styles/variables/variable.scss' ;
```

Mes imports de style après

```
@import '../../../styles/special-theme/colors.scss' ;  
@import '../../../styles/variables/variable.scss' ;
```



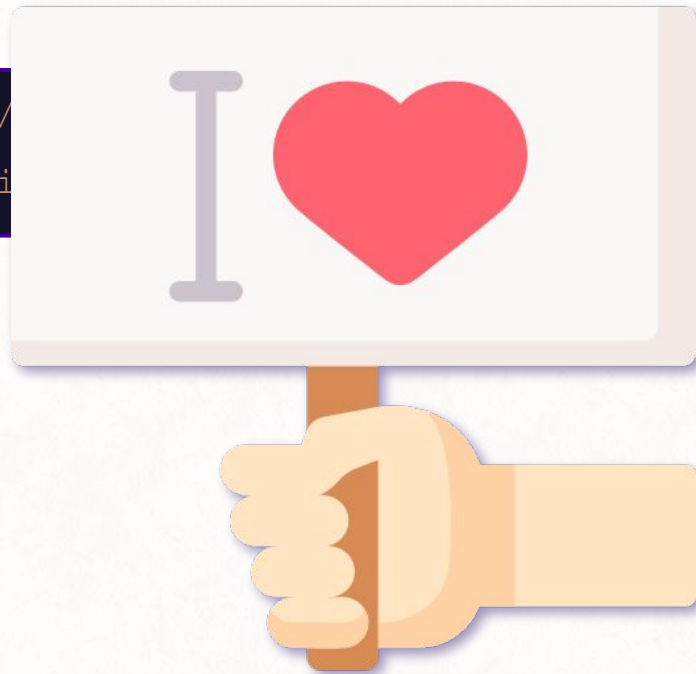
```
@import 'colors';  
@import 'variables';
```

Mes imports de style après

```
@import '../.../styles/special-theme/  
@import '../.../styles/variables/vari
```



```
@import 'colors';  
@import 'variables';
```





TrackBy

Démo pour mettre en vert ce qui est re-render

The screenshot shows a web application running on `localhost:4200`. The page title is "Management de mes étudiants". It displays a list of students with their names and email addresses, each with a "Détails" button. The students listed are:

- Alain Térieur (alaintérieur@gmail.com)
- Alex Térieur (alexterieur@gmail.com)
- Cara Melmou (caramelmou@gmail.com)
- Jean Peupu (jeanpeupu@gmail.com)
- Lara Tatouille (laratatouille@gmail.com)

Below the list is a form to "Sélectionner l'étudiant du mois" with radio buttons for Alain, Alex, Cara, Jean, and Lara. Alex is selected.

On the right, the Chrome DevTools "Rendering" panel is open, showing the "Paint flashing" checkbox checked. This highlights areas of the page (green) that need to be repainted. The panel also shows the "Layout Shift Regions" and "Layer borders" checkboxes.

The video player at the bottom indicates the video is at 0:27 / 1:16.

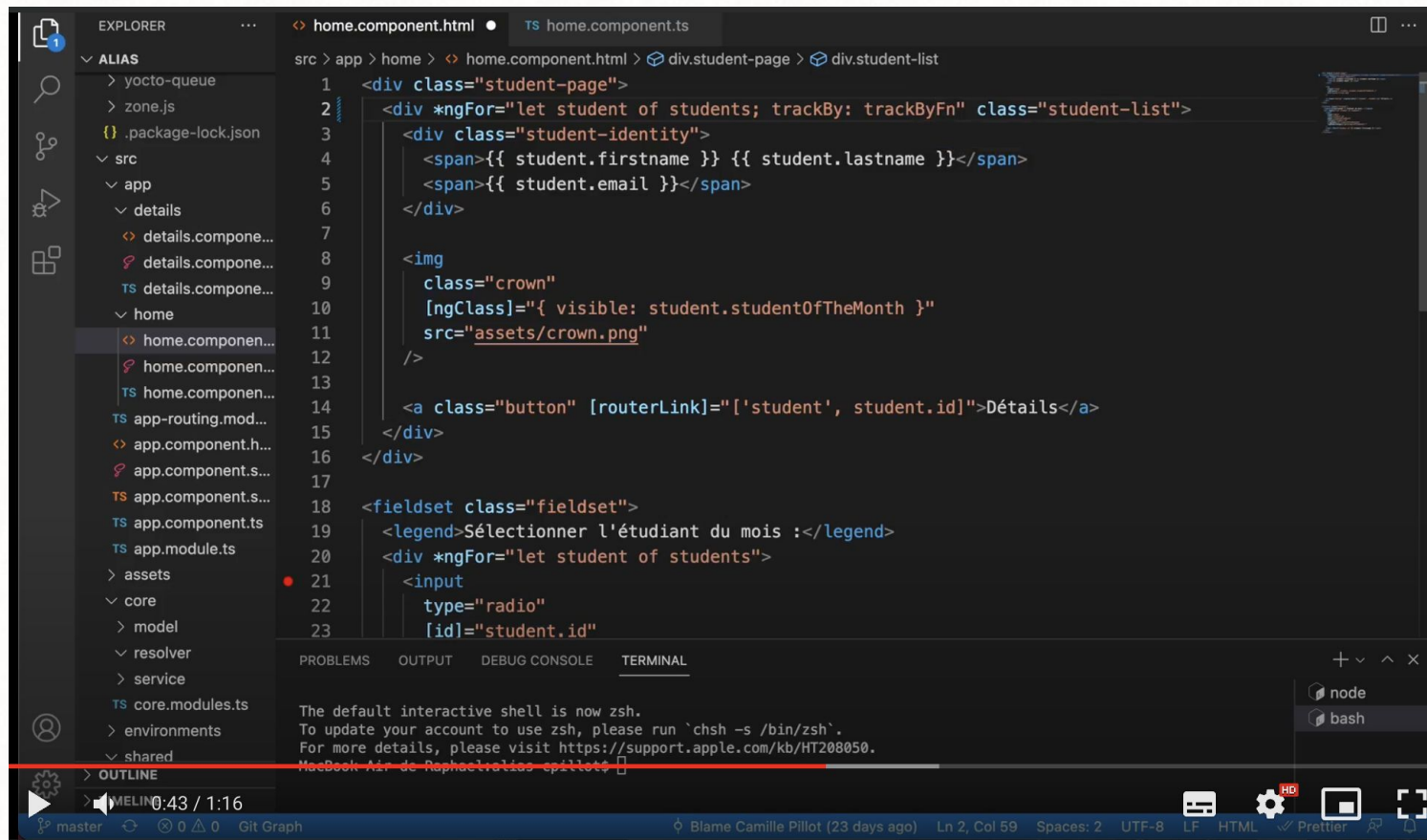
Démo sans trackBy

The screenshot shows a web browser at localhost:4200 displaying a "Management de mes étudiants" application. The application has a pink header and a list of student cards. Each card contains a name, email, and a "Détails" button. A red crown icon is visible between the second and third cards. Below the list is a form to "Sélectionner l'étudiant du mois" with radio buttons for Alain, Alex (selected), Cara, Jean, and Lara.

The browser's developer tools are open on the right, showing the "Elements" panel with the DOM tree. The selected element is a shell component. The "Console" panel shows a "Paint flashing" warning, indicating areas of the page that need to be repainted.

At the bottom of the browser window, there is a video player interface with a play button, a volume icon, and a progress bar showing 0:27 / 1:16.

Démo ajout du trackBy dans le html

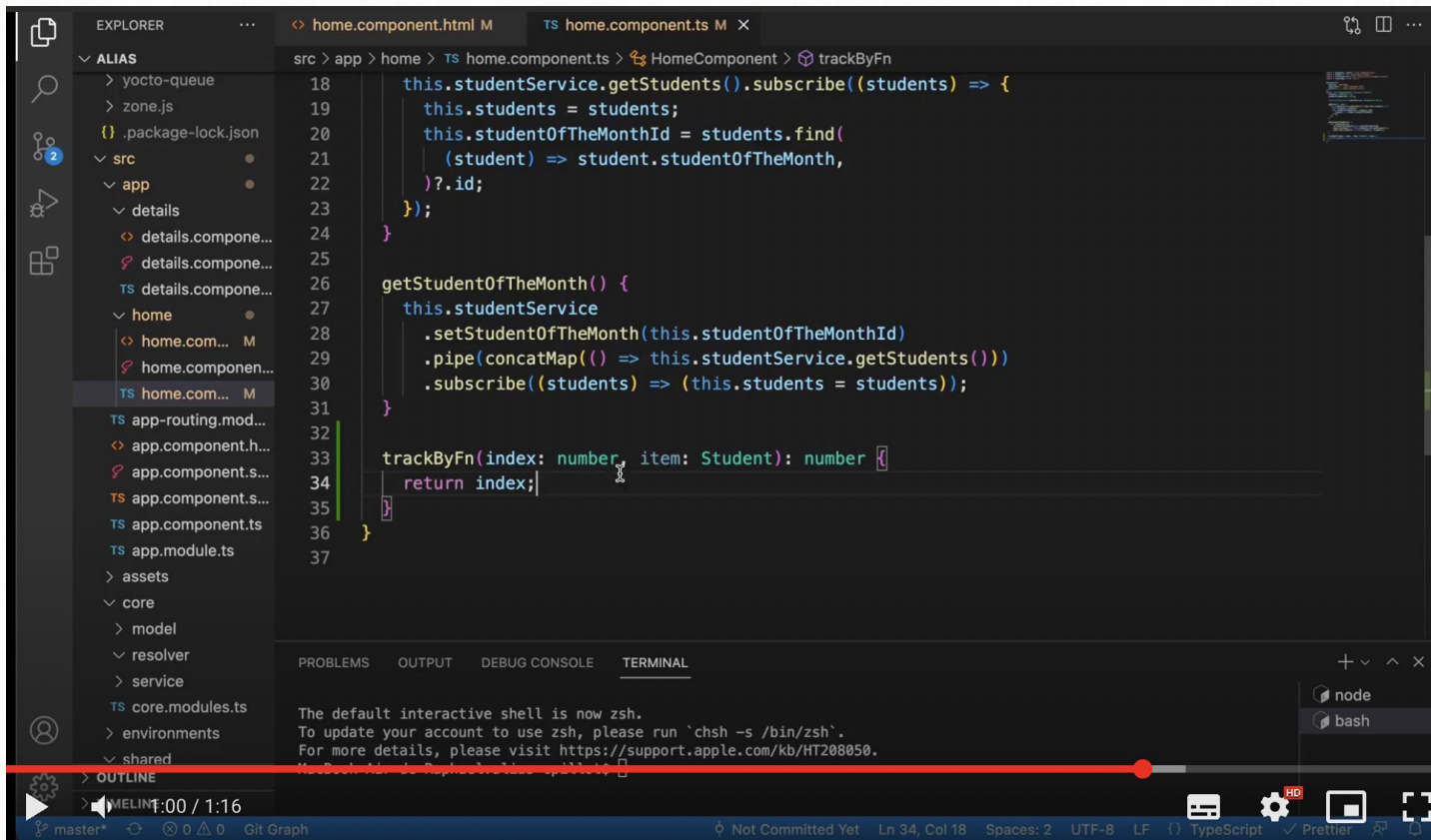


The screenshot shows a VS Code editor with a file explorer on the left and a code editor in the center. The file explorer shows a project structure with a 'home' component. The code editor displays the 'home.component.html' file, which contains an Angular template. The template has two main sections: a list of students and a form to select a student of the month. The first section uses an `*ngFor` loop to iterate over a list of students. The second section uses a `*ngForm` to select a student of the month. The code is as follows:

```
1 <div class="student-page">
2   <div *ngFor="let student of students; trackBy: trackByFn" class="student-list">
3     <div class="student-identity">
4       <span>{{ student.firstname }} {{ student.lastname }}</span>
5       <span>{{ student.email }}</span>
6     </div>
7   </div>
8   
13   <a class="button" [routerLink]="['student', student.id]">Détails</a>
14 </div>
15
16 <fieldset class="fieldset">
17   <legend>Sélectionner l'étudiant du mois </legend>
18   <div *ngForm="let student of students">
19     <input
20       type="radio"
21       [id]="student.id"
22     />
23   </div>
24 </fieldset>
```

The terminal at the bottom shows the default interactive shell is now zsh. To update the account to use zsh, please run `chsh -s /bin/zsh`. For more details, please visit <https://support.apple.com/kb/HT208050>.

Démo ajout de la fonction trackByFn dans le ts



The screenshot shows a VS Code editor with a project structure on the left and a TypeScript file being edited in the center. The Explorer panel on the left shows a project with a 'home' component. The main editor displays the 'home.component.ts' file, which contains a 'trackByFn' function and a 'getStudentOfTheMonth' method. The 'trackByFn' function is defined as follows:

```
trackByFn(index: number, item: Student): number {  
  return index;  
}
```

The 'getStudentOfTheMonth' method is defined as follows:

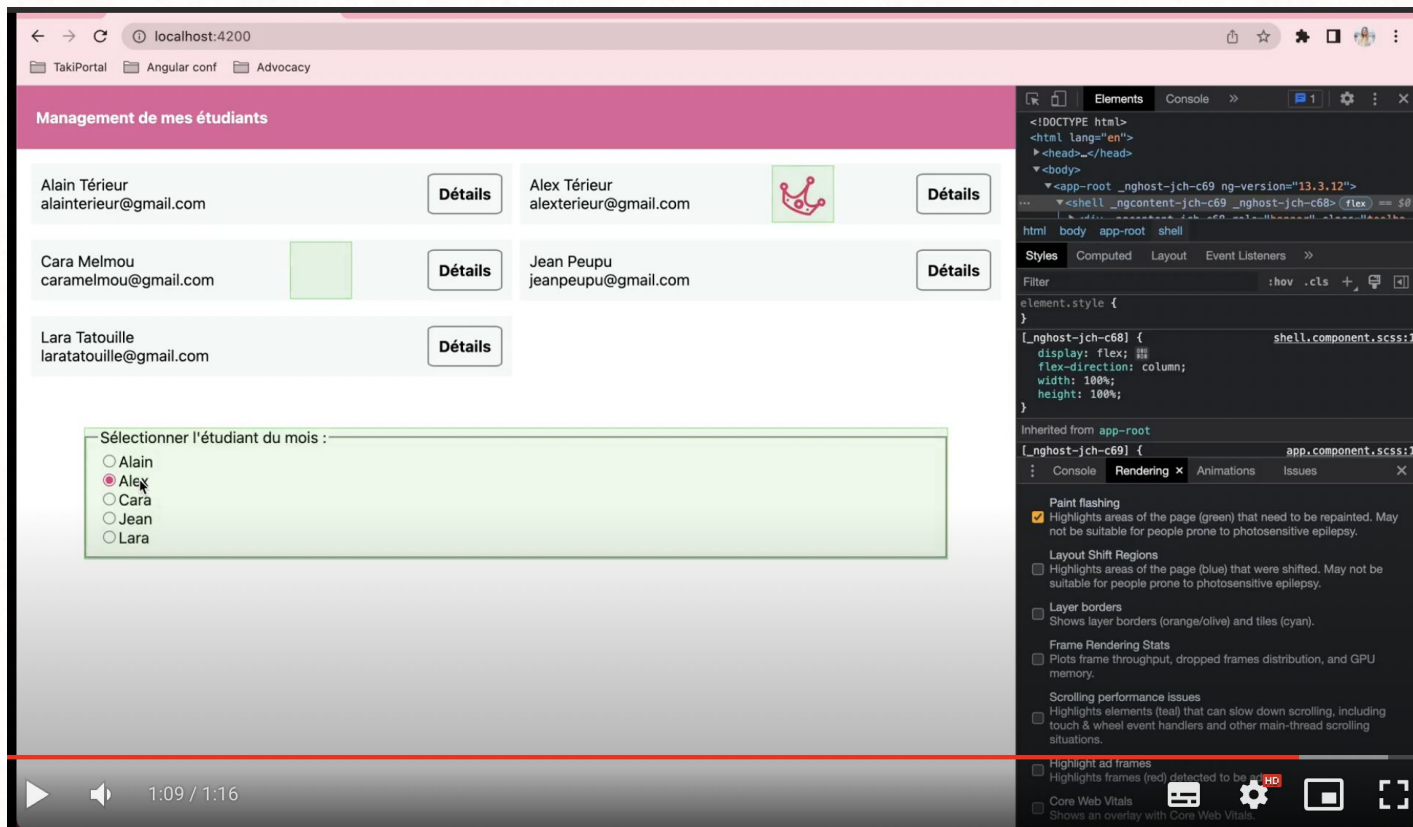
```
getStudentOfTheMonth() {  
  this.studentService  
    .setStudentOfTheMonth(this.studentOfTheMonthId)  
    .pipe(concatMap(() => this.studentService.getStudents()))  
    .subscribe((students) => (this.students = students));  
}
```

The bottom of the editor shows a terminal window with the following text:

```
The default interactive shell is now zsh.  
To update your account to use zsh, please run `chsh -s /bin/zsh`.  
For more details, please visit https://support.apple.com/kb/HT208050.
```

The status bar at the bottom indicates the file is in the 'master' branch, has 0 changes, and is using the 'TypeScript' language.

Démo avec trackBy



The screenshot displays a web application titled "Management de mes étudiants" running on localhost:4200. The application shows a list of students with their names and email addresses. A "Détails" button is next to each student. A dropdown menu is open, showing a list of students to select from.

The students listed are:

- Alain Térieur (alaintérieur@gmail.com)
- Cara Melmou (caramelmou@gmail.com)
- Lara Tatouille (laratatouille@gmail.com)

The dropdown menu is titled "Sélectionner l'étudiant du mois :" and contains the following options:

- ☐ Alain
- ☒ Alex
- ☐ Cara
- ☐ Jean
- ☐ Lara

The right side of the image shows the Chrome DevTools console with the trackBy function defined and the application running. The console output shows the trackBy function being called with the student object and the index.

```
trackBy: function (index, item) {
  return item.id;
}
```

Take Away

Définition



fonction facultative de *ngFor



suivre la modification des éléments dans un itérable

Take Away

Définition



fonction facultative de *ngFor



suivre la modification des éléments dans un itérable

Avantage



performance sur des listes longues

Track

Take Away

Angular 17 : nouveau built-in control flow



*ngFor => @for()



trackBy => track 

```
@for(student of students; track $index){  
  <div class="student-identity">  
    <span>{{ student.firstname }}</span>  
    <span>{{ student.mail }}</span>  
  </div>  
}
```

Avantage



performance sur des listes longues

TrackBy / track Take Away

Avantage



performance sur des listes longues





unsubscribe

Unsubscribe

user.component.ts

```
@Component({
  selector: 'user',
  templateUrl: './user.component.html',
})
export class UserComponent implements OnInit, OnDestroy {
  userLanguage: string;
  subscription: Subscription;
  languageService = inject(LanguageService);

  ngOnInit(): void {
    this.subscription = languageService
      .observeLanguageChange()
      .subscribe((lang) => (this.userLanguage = lang));
  }

  ngOnDestroy(): void {
    this.subscription.unsubscribe();
  }
}
```

user.component.html

```
<div>{{ userLanguage }}</div>
```

Unsubscribe

user.component.ts

```
@Component({
  selector: 'user',
  templateUrl: './user.component.html',
})
export class UserComponent implements OnInit, OnDestroy {
  userLanguage: string;
  subscription: Subscription;
  languageService = inject(LanguageService);

  ngOnInit(): void {
    this.subscription = languageService
      .observeLanguageChange()
      .subscribe((lang) => (this.userLanguage = lang));
  }

  ngOnDestroy(): void {
    this.subscription.unsubscribe();
  }
}
```

user.component.html

```
<div>{{ userLanguage }}</div>
```

Unsubscribe

user.component.ts

```
@Component({
  selector: 'user',
  templateUrl: './user.component.html',
})
export class UserComponent implements OnInit, OnDestroy {
  userLanguage: string;
  subscription: Subscription;
  languageService = inject(LanguageService);

  ngOnInit(): void {
    this.subscription = languageService
      .observeLanguageChange()
      .subscribe((lang) => (this.userLanguage = lang));
  }

  ngOnDestroy(): void {
    this.subscription.unsubscribe();
  }
}
```

user.component.html

```
<div>{{ userLanguage }}</div>
```

Unsubscribe

user.component.ts

```
@Component({
  selector: 'user',
  templateUrl: './user.component.html',
})
export class UserComponent implements OnInit, OnDestroy {
  userLanguage: string;
  subscription: Subscription;
  languageService = inject(LanguageService);

  ngOnInit(): void {
    this.subscription = languageService
      .observeLanguageChange()
      .subscribe((lang) => (this.userLanguage = lang));
  }

  ngOnDestroy(): void {
    this.subscription.unsubscribe();
  }
}
```

user.component.html

```
<div>{{ userLanguage }}</div>
```

Unsubscribe

user.component.ts

```
@Component({
  selector: 'user',
  templateUrl: './user.component.html',
})
export class UserComponent implements OnInit, OnDestroy {
  userLanguage: string;
  subscription: Subscription;
  languageService = inject(LanguageService);

  ngOnInit(): void {
    this.subscription = languageService
      .observeLanguageChange()
      .subscribe((lang) => (this.userLanguage = lang));
  }

  ngOnDestroy(): void {
    this.subscription.unsubscribe();
  }
}
```

user.component.html

```
<div>{{ userLanguage }}</div>
```

Unsubscribe

user.component.ts

```
@Component({
  selector: 'user',
  templateUrl: './user.component.html',
})
export class UserComponent implements OnInit, OnDestroy {
  userLanguage: string;
  subscription: Subscription;
  languageService = inject(LanguageService);

  ngOnInit(): void {
    this.subscription = languageService
      .observeLanguageChange()
      .subscribe((lang) => (this.userLanguage = lang));
  }

  ngOnDestroy(): void {
    this.subscription.unsubscribe();
  }
}
```

user.component.html

```
<div>{{ userLanguage }}</div>
```

Unsubscribe

user.component.ts

```
@Component({
  selector: 'user',
  templateUrl: './user.component.html',
})
export class UserComponent implements OnInit, OnDestroy {
  userLanguage: string;
  subscription: Subscription;
  languageService = inject(LanguageService);

  ngOnInit(): void {
    this.subscription = languageService
      .observeLanguageChange()
      .subscribe((lang) => (this.userLanguage = lang));
  }

  ngOnDestroy(): void {
    this.subscription.unsubscribe();
  }
}
```

user.component.html

```
<div>{{ userLanguage }}</div>
```



Observable = fuite mémoire



Observable = fuite mémoire



les souscriptions subsistent



Observable = fuite mémoire



les souscriptions subsistent



penser à *unsubscribe()*

Unsubscribe



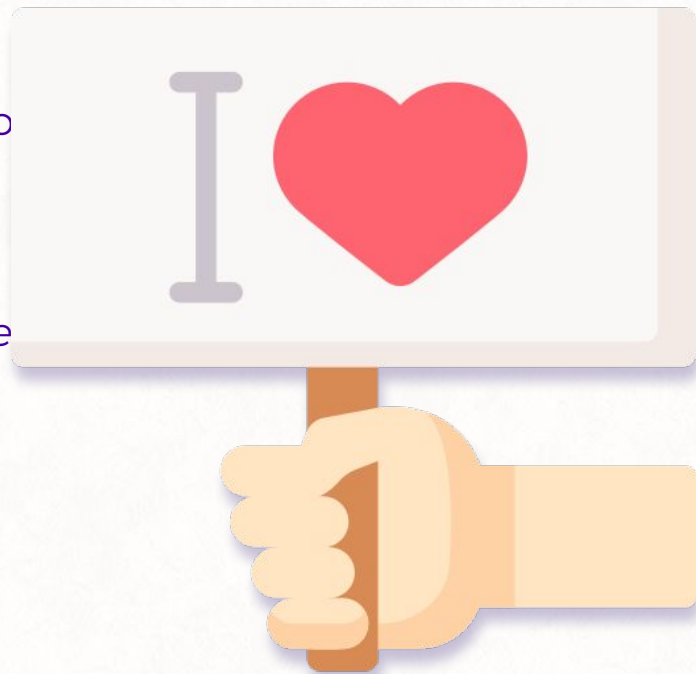
Observable = fuite mémoire



les souscriptions subsistent

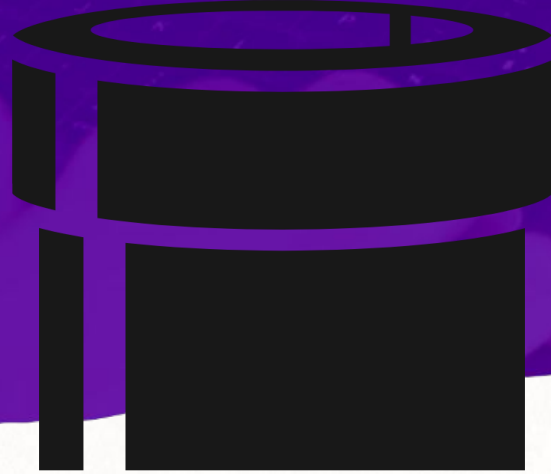


penser à unsubscribe





Le pipe



Le pipe

Qu'est ce que c'est ?

Définition



fonction exclusivement côté HTML



une entrée



une sortie transformée

```
<div>{{ user.name | uppercase }} </div>
```

Le pipe

Qu'est ce que c'est ?

Définition

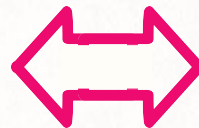


fonction exclusivement côté HTML

```
<div>{{ user.name | uppercase }} </div>
```



une entrée



une sortie transformée

```
function uppercase(username: string): string {  
  return username.toUpperCase()  
}
```

Le pipe

Qu'est ce que c'est ?

Définition

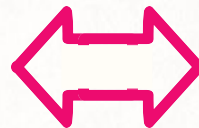


fonction exclusivement côté HTML

```
<div>{{ user.name | uppercase }} </div>
```



une entrée



une sortie transformée

```
function uppercase(username: string): string {  
  return username.toUpperCase()  
}
```



dans *CommonModule* : uppercase mais pas que...



lowercase



lowercase



date



lowercase



date



currency



lowercase



date



decimal



currency



lowercase



date



async



decimal



currency

Les pipes | async



utilisable sur des observables



subscribe implicitement

user.component.ts

```
user$: Observable<User>;
```

user.component.html

```
<div>{{ user$ | async }}</div>
```



pas besoin de *unsubscribe*

user.component.ts

```
export class UserComponent implements OnInit, OnDestroy {  
  userLanguage: string;  
  subscription: Subscription;  
  languageService = inject(LanguageService);  
  
  ngOnInit(): void {  
    this.subscription = languageService  
      .observeLanguageChange()  
      .subscribe((lang) => (this.userLanguage = lang));  
  }  
  
  ngOnDestroy(): void {  
    this.subscription.unsubscribe();  
  }  
}
```

user.component.html

```
<div>{{ userLanguage }}</div>
```

user.component.ts

```
export class UserComponent {  
  languageService = inject(LanguageService);  
  userLanguage$ = this.languageService  
    .observeLanguageChange();  
}
```

user.component.html

```
<div>{{ userLanguage$ | async }}</div>
```

user.component.ts

```
export class UserComponent {  
  languageService = inject(LanguageService);  
  userLanguage$ = this.languageService  
    .observeLanguageChange();  
}
```

user.component.html

```
<div>{{ userLanguage$ | async }}</div>
```

user.component.ts

```
export class UserComponent {  
  languageService = inject(LanguageService);  
  userLanguage$ = this.languageService  
    .observeLanguageChange();  
}
```

user.component.html

```
<div>{{ userLanguage$ | async }}</div>
```

user.component.ts

```
export class UserComponent {  
  languageService = inject(LanguageService);  
  userLanguage$ = this.languageService  
    .observeLanguageChange();  
}
```

user.component.html

```
<div>{{ userLanguage$ | async }}</div>
```



Utiliser le | async

user.component.ts

```
user$: Observable<User>;
```

user.component.html

```
<span>(user$ | async)?.firstname</span>  
<span>(user$ | async)?.lastname</span>
```

Utiliser le | async

user.component.ts

```
user$: Observable<User>;
```

user.component.html

```
<span>(user$ | async)?.firstname</span>  
<span>(user$ | async)?.lastname</span>
```



2 souscriptions créées

Utiliser le | async avec *ngIf as

user.component.ts

```
user$: Observable<User>;
```

user.component.html

```
<ng-container *ngIf="user$ | async as user">  
  <span>{{ user.firstname }}</span>  
  <span>{{ user.lastname }}</span>  
</ng-container>
```

Les pipes

Utiliser le | async avec *ngIf as

user.component.ts

```
user$: Observable<User>;
```

user.component.html

```
<ng-container *ngIf="user$ | async as user">  
  <span>{{ user.firstname }}</span>  
  <span>{{ user.lastname }}</span>  
</ng-container>
```



souscription unique

Les pipes

Utiliser le | async avec @if as

Angular 17 : nouveau built-in control flow

user.component.ts

```
user$: Observable<User>;
```

user.component.html

```
@if(user$ | async; as user){  
  <span>{{ user.firstname }}</span>  
  <span>{{ user.lastname }}</span>  
}
```



souscription unique

Les pipes

Utiliser le | async avec *ngIf/@if as

user.component.ts

```
user$: Observable<User>;
```

```
<ng-container *ngIf="user$ | async">  
  <span>{{ user.f }}</span>  
  <span>{{ user.l }}</span>  
</ng-container>
```

use



souscription unique



Les Resolvers

Resolver

Définition



type de service
ou fonction



résoudre des données

Resolver

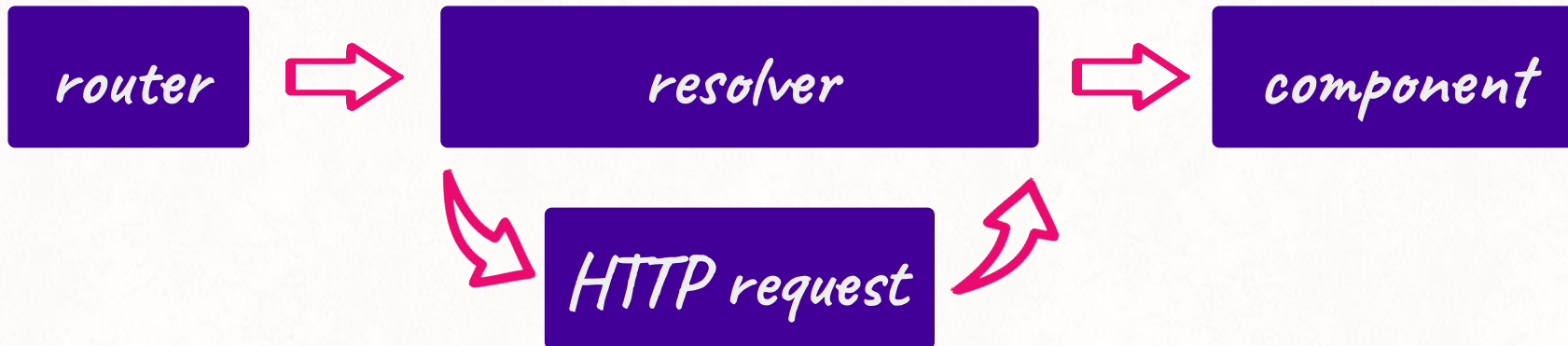
Définition



type de service
ou fonction



résoudre des données



Resolver

Démo en slow 3G

The screenshot shows a web browser at `localhost:4200` displaying a web application titled "Management de mes étudiants". The application lists three students, each with a "Détails" button:

- Alain Térieur (alaintérieur@gmail.com)
- Alex Térieur (alexterieur@gmail.com)
- Cara Melmou (caramelmou@gmail.com)
- Jean Peupu (jeanpeupu@gmail.com)
- Lara Tatouille (laratatouille@gmail.com)

Below the list, there is a section "Sélectionner l'étudiant du mois :" with radio buttons for Alain, Alex, Cara, Jean, and Lara. Alain is selected.

The Chrome DevTools Network panel is open on the right, showing a list of requests. The "No throttling" dropdown is open, and "Slow 3G" is selected. The "Presets" dropdown is also open, showing "Fast 3G", "Slow 3G", and "Offline". The "Slow 3G" option is highlighted.

The video player at the bottom shows a progress bar at 0:10 / 3:14.

Resolver

Démo sans resolver

The screenshot shows a web browser at the URL `localhost:4200/student/4611b784-d091-4b37-a05f-f8bdda508597`. The page title is "Management de mes étudiants". A "Retour" link is visible. A large pink loading spinner is centered on the page with the text "Chargement..." below it. The browser's developer tools are open to the Network tab, showing a list of requests. The requests are as follows:

Name	S.	T.	Init..	S.	T.	Waterfall
localh...	3...	d...	Ot...	2...	2...	
css2?f...	2...	s...	(in...	(...	0...	
icon?f...	2...	s...	(in...	(...	0...	
KFOm...	2...	f...	cs...	(...	0...	
styles....	3...	s...	(in...	2...	2...	
runtim...	3...	s...	(in...	2...	2...	
polyfill...	3...	s...	(in...	2...	2...	
vendor.js	3...	s...	(in...	2...	2...	
main.js	3...	s...	(in...	2...	2...	
styles.js	3...	s...	(in...	2...	2...	
students	2...	x...	zo...	1...	1...	
ws	1...	w...	We...	0...	P...	
crown...	3...	p...	pla...	2...	3...	
4611b...	(...	x...	zo...	0...	P...	
details...	(...	x...	zo...	0...	P...	
back-...	(...		pla...	0...	P...	

The bottom of the browser shows a video player interface with a progress bar at 0:17 / 3:14 and a volume icon.

Resolver

Démo sans resolver

localhost:4200/student/4611b784-d091-4b37-a05f-f8bdda508597

TakiPortal Angular conf Advocacy

Management de mes étudiants

Retour

Prénom : Jean
Nom : Peupu
Age : 30 ans
Mail: jeanpeupu@gmail.com

Formation Java

Formation JavaScript

Formation TypeScript

Bonjour Camille, j'ai fini la formation Java !

Ecrire un message...

Envoyer

0:28

0:27 / 3:14

67

Démo ajout d'un functional resolver

```
MacBook-Air-de-Raphael:resolver cpilot$ ng generate resolver student
CREATE src/core/resolver/student.resolver.spec.ts (370 bytes)
CREATE src/core/resolver/student.resolver.ts (402 bytes)
MacBook-Air-de-Raphael:resolver cpilot$
```

▼ TIPS

TS app.component.ts

TS app.config.ts

TS app.routes.ts M

> assets

▼ core ●

> model

▼ resolver ●

TS student.re... U

TS student.re... U

> service

> shared

★ favicon.ico

<> index.html

tips > src > core > resolver > TS student.resolver.ts > ...

1

▼ import { inject } from '@angular/core';

2

import { ResolveFn } from '@angular/router';

3

import { Student } from '@core/model/student';

4

import { StudentService } from '@core/service/student.service';

5

6

▼ export const studentResolver: ResolveFn<Student> = (route) => {

7

const studentService = inject(StudentService);

8

const id = route.params['id'];

9

return studentService.getStudent(id);

10

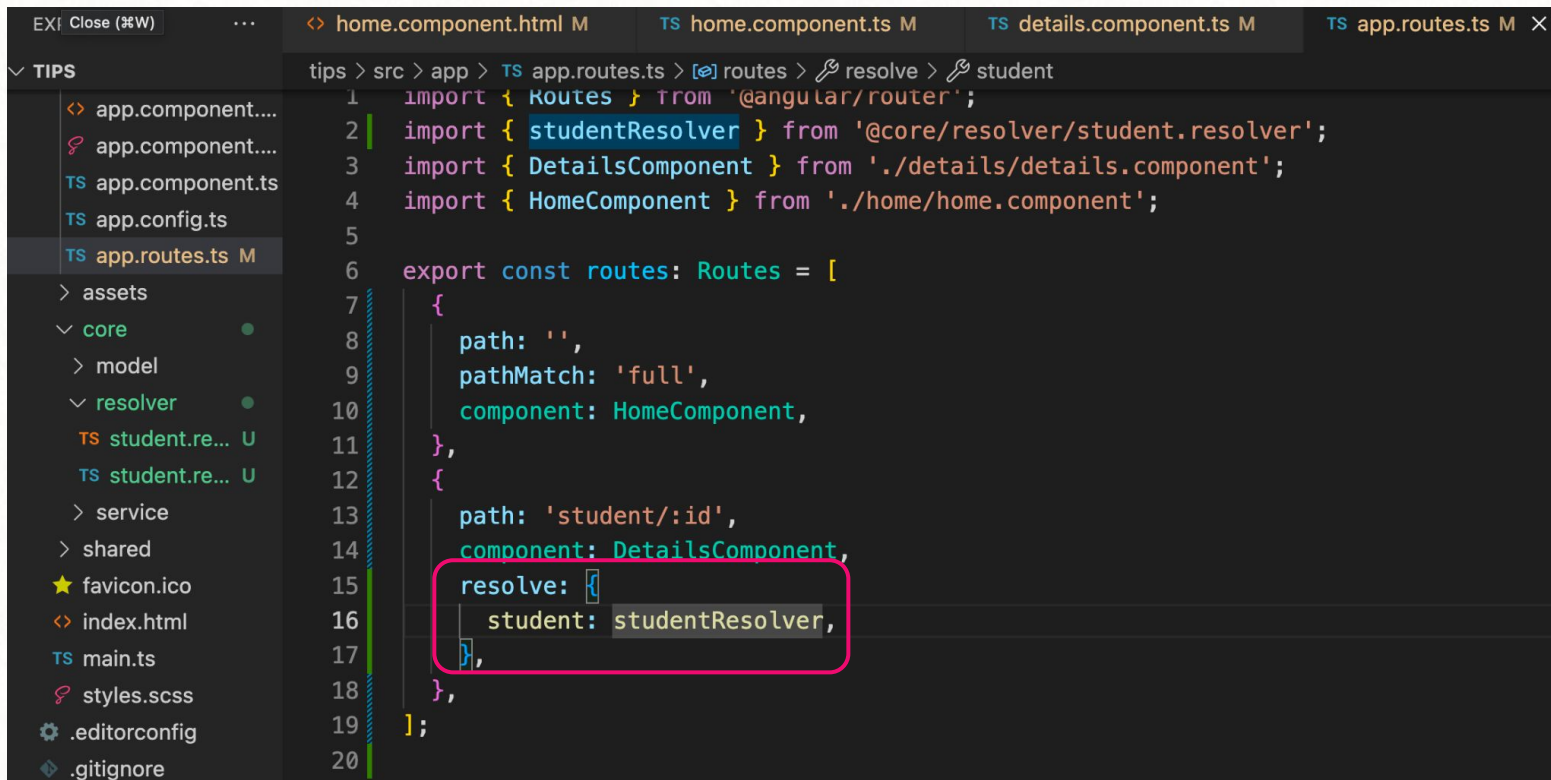
};

11

|

Resolver

Démo ajout resolver sur la route



The screenshot shows an IDE with a file explorer on the left and a code editor on the right. The file explorer shows a project structure with a 'resolver' folder containing 'student.resolver.ts'. The code editor shows the 'app.routes.ts' file with the following code:

```
1 import { Routes } from '@angular/router';
2 import { studentResolver } from '@core/resolver/student.resolver';
3 import { DetailsComponent } from './details/details.component';
4 import { HomeComponent } from './home/home.component';
5
6 export const routes: Routes = [
7   {
8     path: '',
9     pathMatch: 'full',
10    component: HomeComponent,
11  },
12  {
13    path: 'student/:id',
14    component: DetailsComponent,
15    resolve: {
16      student: studentResolver,
17    },
18  },
19 ];
```

The 'resolve' property in the second route is highlighted with a red box, indicating the addition of the 'student' resolver.

Resolver

Démo avant utilisation resolver dans le composant

```
EXPLORER
  ALIAS
    app
      details
        details.compone...
        details.compone...
        TS details.compone...
      home
        home.com... M
        home.componen...
        TS home.com... M
        TS app-routing.... M
        app.component.h...
        app.component.s...
        TS app.component.s...
        TS app.component.ts
        TS app.module.ts
    assets
    core
    model
    resolver
      TS student.res... U
      TS student.res... U
    service
    TS core.modules.ts
    environments
    shared
    components
    banner
  OUTLINE
  2:35 / 3:14

TS details.component.ts
src > app > details > TS details.component.ts > DetailsComponent > ngOnInit > studentId
10 selector: 'app-details',
11 templateUrl: './details.component.html',
12 styleUrls: ['./details.component.scss'],
13 })
14 export class DetailsComponent implements OnInit {
15   student$: Observable<Student>;
16   details$: Observable<Details>;
17
18   constructor(
19     private route: ActivatedRoute,
20     private studentService: StudentService,
21     private detailsService: DetailsService,
22   ) {}
23
24   ngOnInit(): void {
25     const studentId = this.route.snapshot.params['id'];
26     this: this
27     this.student$ = this.studentService.getStudent(studentId);
28     this.details$ = this.detailsService.getDetailsFromStudent(studentId);
29   }
30 }
31
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
MacBook-Air-de-Raphael:resolver cpillot$
MacBook-Air-de-Raphael:resolver cpillot$
MacBook-Air-de-Raphael:resolver cpillot$
MacBook-Air-de-Raphael:resolver cpillot$ ng generate resolver student
CREATE src/core/resolver/student.resolver.spec.ts (370 bytes)
CREATE src/core/resolver/student.resolver.ts (402 bytes)
MacBook-Air-de-Raphael:resolver cpillot$
```

node
bash resolv...

Blame Camille Pillot (23 days ago) Ln 25, Col 19 Spaces: 2 UTF-8 LF TypeScript Prettier

Resolver

Démo après utilisation resolver dans le composant

```
src > app > details > TS details.component.ts > DetailsComponent > ngOnInit
10 selector: 'app-details',
11 templateUrl: './details.component.html',
12 styleUrls: ['./details.component.scss'],
13 })
14 export class DetailsComponent implements OnInit {
15     student$: Observable<Student>;
16     details$: Observable<Details>;
17
18     constructor(
19         private route: ActivatedRoute,
20         private studentService: StudentService,
21         private detailsService: DetailsService,
22     ) {}
23
24     ngOnInit(): void {
25         const studentId = this.route.snapshot.params['id'];
26
27         this.student$ = this.route.data.pipe(map((data) => data['student']));
28         this.details$ = this.detailsService.getDetailsFromStudent(studentId);
29     }
30 }
31
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL

```
MacBook-Air-de-Raphael:resolver cpillot$
MacBook-Air-de-Raphael:resolver cpillot$
MacBook-Air-de-Raphael:resolver cpillot$
MacBook-Air-de-Raphael:resolver cpillot$ ng generate resolver student
CREATE src/core/resolver/student.resolver.spec.ts (370 bytes)
CREATE src/core/resolver/student.resolver.ts (402 bytes)
MacBook-Air-de-Raphael:resolver cpillot$
```

2:46 / 3:14

Blame Camille Pillot (23 days ago) Ln 27, Col 74 Spaces: 2 UTF-8 LF TypeScript Prettier

Resolver

Démo avec resolver données directement affichées

The screenshot shows a web browser at the URL `localhost:4200/student/4611b784-d091-4b37-a05f-f8bdda508597`. The page title is "Management de mes étudiants". It features a "Retour" link and a student profile card for Jean Peupu, 30 years old, with email `jeanpeupu@gmail.com`. A loading spinner and the text "Chargement..." are visible on the right side of the page.

The browser's developer tools are open to the Network tab, displaying a list of requests. The following table summarizes the visible requests:

Name	S.	T.	Init..	S.	T.	Waterfall
localh...	3...	d...	2...	2...		
css2?f...	2...	s...	(in...	(...	3...	
icon?f...	2...	s...	(in...	(...	3...	
styles...	3...	s...	(in...	2...	2...	
runtim...	2...	s...	(in...	7...	2...	
polyfill...	3...	s...	(in...	2...	2...	
vendor.js	3...	s...	(in...	2...	2...	
main.js	2...	s...	(in...	5...	3...	
styles.js	3...	s...	(in...	2...	2...	
KFOm...	2...	f...	cs...	(...	0...	
ws	1...	w...	We...	0...	2...	
students	2...	x...	zo...	1...	2...	
favico...	3...	v...	Ot...	2...	2...	
crown...	3...	p...	pla...	2...	2...	
ws	1...	w...	We...	0...	2...	
ws	1...	w...	We...	0...	2...	
4611b...	2...	x...	zo...	4...	2...	
details...	(...	x...	zo...	0...	P...	
back...	(...	p...	pla...	0...	P...	
boy.png	(...	p...	pla...	0...	P...	

Resolver

Désavantage



Le *resolver* empêche l'affichage
TOTAL de la page

Resolver

Exemple OVH

Gros temps de chargement puis toutes les informations

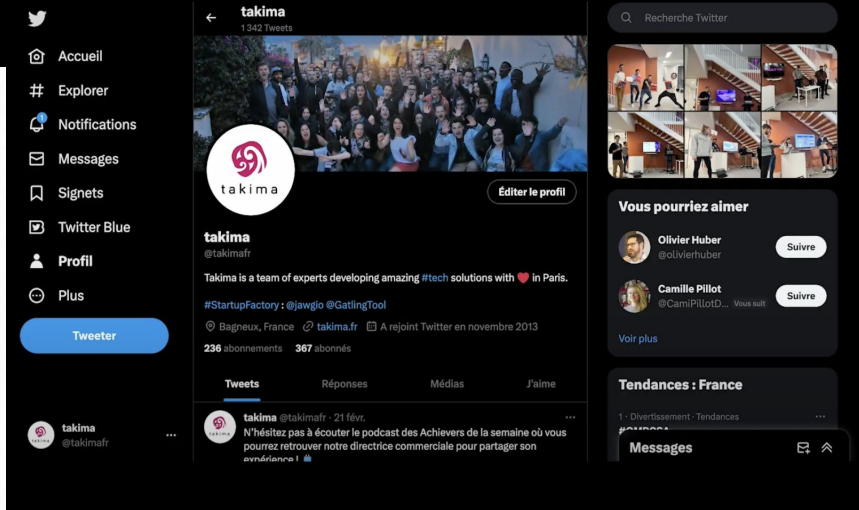
The screenshot displays the OVHcloud Public Cloud dashboard. The top navigation bar includes links for 'Tableau de bord', 'Bare Metal Cloud', 'Hosted Private Cloud', 'Public Cloud' (selected), 'Web Cloud', 'Télécom', 'Sunrise', and 'Marketplace'. The left sidebar lists categories: 'Compute' (with a '+ Créer un nouveau projet' button), 'Storage', 'Network', and 'Containers & Orchestration'. The main content area features a grid of service tiles: 'Créer une instance', 'Créer un cluster Kubernetes', 'Créer un conteneur Object Storage', 'Créer un volume de Block Storage', 'Créer une base de données', 'Créer un AI Notebook', and 'Gérer les réseaux privés (vRack)'. Below this grid are three sections: 'Facturation' (showing a 0.00 € balance), 'Documentation' (with links to Getting Started, Public Cloud, Instances, and Facturation), and 'Communauté' (with links to Roadmap and Communauté).



Chargement en cours...

Resolver

Exemple Twitter

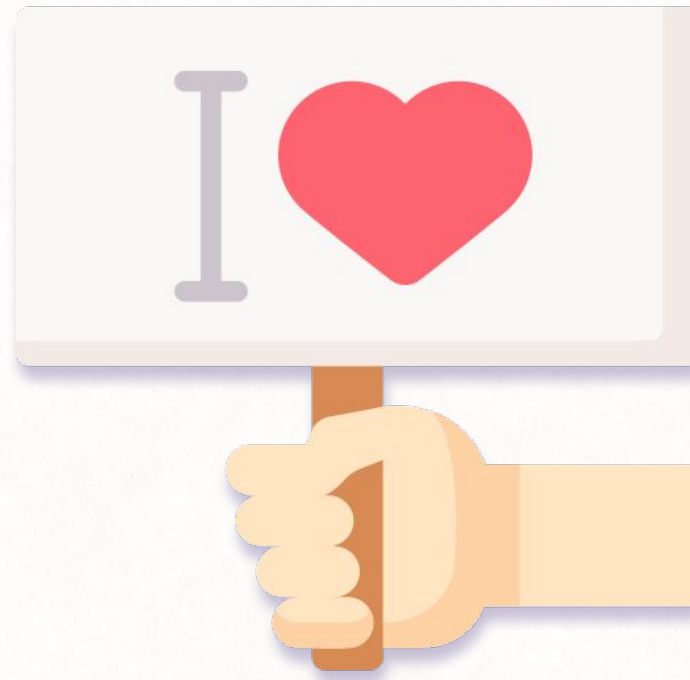


Petit temps de chargement,
informations importantes
présentes de suite puis le
reste se load
progressivement

Expérience utilisateur



Resolver + loader



A photograph of a person's hands typing on a laptop keyboard, overlaid with a semi-transparent purple filter. The text "Pour résumer..." is written in a white, cursive font across the center of the image.

Pour résumer...

Les alias

```
import { Banner } from  
    '../../../core/models/model' ;
```



```
import { Banner } from '@core/models/model' ;
```

trackBy

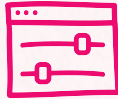


```
<div *ngFor="let user of users; trackBy: trackByFn">  
  <span>{{ user.firstname }}</span>  
</div>
```

v17+

```
@for(user of users; track user.id){  
  <span>{{ user.firstname }}</span>  
}
```

Outils navigateur



paint flashing



limitation de bande
passante

unsubscribe



`.subscribe()`



`.unsubscribe()`





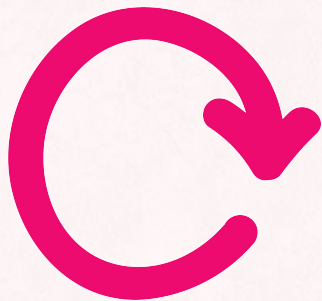
| async

*ngIf as

@if() as

v17+

Resolvers



précharge les données



à utiliser selon le cas d'usage

Conclusion

Les tips découverts

Merci !



Camille Pillot



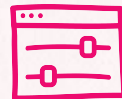
@CamiPillotDev



alias



trackBy



outils navigateur



unsubscribe



| async



resolvers